

HKUST MATH 2431 - problem set 4

1.

1. (*) Benford's law and σ -algebras

[4 Points]

Empirical evidence suggests that the first digits in lists of large data sets of real numbers appearing naturally (such as natural constants, lengths of rivers, stock prices, ...) are *not* uniformly distributed on $\{1, \dots, 9\}$. Instead, they follow the so-called Benford distribution, which is a discrete distribution on $\{1, \dots, 9\}$ with

$$p(k) = \log_{10} \left(1 + \frac{1}{k} \right), \quad k \in \{1, \dots, 9\}$$

(Newcomb 1881, Benford 1938). We want to make this mathematically plausible, and aim at constructing a probability measure $\mathbf{P}$ on $(0, \infty)$ equipped with an appropriate σ -algebra $\mathcal{M}$ with

$$(\star) \quad \mathbf{P}[\{\omega \in (0, \infty) : \text{the decimal representation of } \omega \text{ starts with digit } k\}] = \log_{10} \left(1 + \frac{1}{k} \right).$$

It is clear that $\mathbf{P}$ has to be scale invariant for all $\alpha > 0$, i.e. $\mathbf{P}[\alpha B] = \mathbf{P}[B]$ for all $\alpha > 0$ and $B \in \mathcal{M}$. One can show that for $\mathcal{B}((0, \infty)) = \{B \cap (0, \infty) : B \in \mathcal{B}(\mathbb{R})\}$, no such probability measure exists. However, one can consider a smaller σ -algebra

$$\mathcal{M} = \sigma(\mathcal{E}), \text{ where } \mathcal{E} = \{M_t : 1 < t < 10\}, \text{ with sets } M_t = \bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, t).$$

One can then show that the expression $\mathbf{P}[M_t] := \log_{10}(t)$ can be extended to a probability measure $\mathbf{P}$ on $((0, \infty), \mathcal{M})$ (you do not need to show this).

(a) Show that $\alpha M_t \in \mathcal{M}$ for all $\alpha > 0$ and all $M_t \in \mathcal{E}$.

1. cr. $\alpha > 0$ can always be written as

$$\alpha = \beta \cdot 10^n, \quad 1 \leq \beta < 10, \quad n \in \mathbb{Z}$$

We prove that ① $10^n M_t \in \mathcal{M}$ for all $n \in \mathbb{Z}$ and $t \in (1, 10)$

and ② $\beta M_t \in \mathcal{M}$ for all $\beta \in [1, 10)$ and $t \in (1, 10)$

then $\alpha M_t \in \mathcal{M}$ follows from $M_t \in \mathcal{M} \Rightarrow 10^n M_t = M_t \in \mathcal{M}$ (by ①)

$$\begin{aligned} \text{For ①, } 10^n M_t &= 10^n \bigcup_{m \in \mathbb{Z}} 10^m \cdot [1, t) \stackrel{?}{=} \alpha M_t = \beta \cdot 10^n M_t = \beta M_t \in \mathcal{M} \\ &= \bigcup_{m \in \mathbb{Z}} 10^{n+m} \cdot [1, t) \quad (\text{by ②}) \\ &= \bigcup_{n' \in \mathbb{Z}} 10^{n'} \cdot [1, t) \quad \text{because } m \mapsto n+m \text{ is bijective on } \mathbb{Z} \\ &= M_t \in \mathcal{M} \end{aligned}$$

For ② either $\beta t \in (1, 10)$ or $\beta t \in [10, 100)$

When $\beta t \in (1, 10)$, $\beta M_t = \beta \bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, t)$

(Note for convenience, we additionally define

$M_1 = \emptyset$, this is used when $\beta = 1$)

$$= \bigcup_{n \in \mathbb{Z}} 10^n \cdot [\beta, \beta t)$$

$$= \left(\bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, \beta t) \right) - \left(\bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, \beta) \right)$$

$$= M_{\beta t} - M_{\beta} = M_{\beta t} \cap (M_{\beta})^c$$

$$\Rightarrow \beta M_t \in \mathcal{M}$$

when $\beta t \in [10, 100)$, $\beta M_t = \beta \bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, t)$

$$= \bigcup_{n \in \mathbb{Z}} 10^n \cdot [\beta, \beta t)$$

Similarly, for this case,
it is used when $\beta_t = 10$

$$= \left(\bigcup_{n \in \mathbb{Z}} 10^n \cdot [\beta, 10) \right) \cup \left(\bigcup_{n \in \mathbb{Z}} 10^n \cdot [1, \frac{\beta}{10}) \right)$$

note $n \mapsto n+1$ is bijective
on $\mathbb{Z}$

$$= (M_\beta)^c \cup M_{\frac{\beta}{10}}$$

$$\Rightarrow \beta M_t \in \mathcal{M}$$

$$\text{So } \alpha M_t = \beta \cdot 10^v M_t \in \mathcal{M}.$$

(b) Show that $\mathbf{P}[\alpha M_t] = \mathbf{P}[M_t]$.

1.b. Some proofs in 1.a. can be reused.

Again, $\alpha \equiv \beta \cdot 10^n$ $\beta \in [1, 10)$, $n \in \mathbb{Z}$, and split by ① and ②

For ①, $10^n M_t = M_t \Rightarrow P[10^n M_t] = P[M_t]$

For ②, when $\beta t \in [1, 10)$
 $\beta M_t = M_{\beta t} \cap (M_\beta)^c$

$$= (M_{\beta t})^c \cup M_\beta$$

$$P[\beta M_t] = 1 - \left((1 - \log_{10}(\beta t)) + \log_{10}(\beta) t \right) + \log_{10}\left(\frac{\beta t}{10}\right)$$

$$= \log_{10}(\beta t) - \log_{10}(\beta)$$

$$= \log_{10}(t)$$

$$= P[M_t]$$

when $\beta t \in [10, 100)$
 $\beta M_t = (M_\beta)^c \cup M_{\frac{\beta t}{10}}$
 $P[\beta M_t] = (1 - \log_{10}(\beta)) + \log_{10}\left(\frac{\beta t}{10}\right)$

no overlap between $(M_{\beta t})^c$ and M_β because $M_\beta \subseteq M_{\beta t}$ as $\beta \leq \beta t$

no overlap because $M_{\frac{\beta t}{10}} \subseteq M_\beta$ as $\frac{\beta t}{10} < \beta$.

$$= \log_{10}(t)$$

$$= P[M_t]$$

So $P[\alpha M_t] = P[\underbrace{\beta \cdot 10^n}_{= M_t} M_t] = P[\beta \cdot M_t] = P[M_t]$

(c) Show that (*) holds.

i.e. $\{\omega \in (0, \infty) : \text{dec. repr of } \omega \text{ starts with } k\}$

that is, $\omega = t \cdot 10^n$ where $t \in [k, k+1)$
for all $n \in \mathbb{Z}$

It is the event $M_{k+1} - M_k \in \mathcal{M}$

(defining $M_1 = \emptyset$ for convenience)

$$\begin{aligned} P[M_{k+1} - M_k] &= P[M_{k+1}] - P[M_k] \quad M_k \subset M_{k+1} \\ &= \log(k+1) - \log(k) = \log\left(1 + \frac{1}{k}\right) \quad k \in \{1, \dots, 8\} \end{aligned}$$

Additionally,

$$\text{defining } M_{10} = \bigcup_{n \in \mathbb{N}_{\neq 0}} M_{10 - \frac{9}{n}} \quad M_{10} \in \mathcal{M} \text{ as a countable union of } M_t$$

$$\begin{aligned} P[M_{10}] &= P\left[\lim_{n \rightarrow \infty} M_{10 - \frac{9}{n}}\right] \quad \text{as } M_{10 - \frac{9}{n}} \subset M_{10 - \frac{9}{m}} \text{ for } n < m. \\ &= \lim_{n \rightarrow \infty} \log\left(10 - \frac{9}{n}\right) \\ &= 1 \end{aligned}$$

(this shows (*) is valid for $k=9$ as well)

(d) Show that the properties of (a) and (b) hold for all $M \in \mathcal{M}$.

Hint: Show that $\mathcal{M}' = \{M \in \mathcal{M} : \alpha M \in \mathcal{M}\}$ is a σ -algebra. You can use without proof that $\mathcal{M}'' = \{M \in \mathcal{M} : \mathbf{P}[\alpha M] = \mathbf{P}[M]\}$ also a σ -algebra. Why does this imply the result?

1. d. For M' ?

① $(0, \infty) \in M'$

- $(0, \infty) \in M$ is true
- $\alpha(0, \infty) \in M$ is true because $\alpha(0, \infty) = (0, \infty)$

② For same $M \in M'$, $M^c \in M'$

- $M^c \in M$ is true because $M \in M$ and M is σ -algebra

- $\alpha M^c \in M$ is true

$$\alpha M^c = (\alpha M)^c \quad x \in \alpha M^c \Rightarrow \frac{x}{\alpha} \in M^c$$

$$(\alpha M)^c \in M$$

$$\Leftrightarrow \alpha M \in M$$

$$\Rightarrow \frac{x}{\alpha} \notin M$$

$$\Rightarrow x \notin \alpha M$$

$$\Rightarrow x \in (\alpha M)^c$$

$\alpha M \in M$ is true

$$\text{because } \alpha M = \alpha \left(\bigcup_{i \in \mathbb{N}} M_{t_i} \right)$$

$$= \bigcup_{i \in \mathbb{N}} \alpha M_{t_i}$$

(*)

but we know $\alpha M_{t_i} \in M$

③ $\left(\bigcup_{i \in \mathbb{N}} M_i \right) \in M'$, where $M_i \in M'$

- $\left(\bigcup_{i \in \mathbb{N}} M_i \right) \in M$ is true because M is σ -alg

- $\alpha \left(\bigcup_{i \in \mathbb{N}} M_i \right) \in M$ is true because

$$\alpha \left(\bigcup_{i \in \mathbb{N}} M_i \right) = \bigcup_{i \in \mathbb{N}} \alpha M_i \quad M_i \in M$$

$$\Rightarrow \alpha M_i \in M \text{ by } (*)$$

$\in M$ as a countable union of elements in M .

So $\mathcal{M}'$ is a σ -alg.

Given $\mathcal{M}''$ is a σ -alg

Note $M_t \in \mathcal{M}'$ and $M_t \in \mathcal{M}''$ for $t \in (1, 10)$ ^{all}

And $\mathcal{M}', \mathcal{M}'' \subseteq \mathcal{M}$,

but $\mathcal{M}$ is the smallest σ -alg containing M_t ,

so in fact, $\mathcal{M}' = \mathcal{M}'' = \mathcal{M}$

Since $\mathcal{M}'' = \mathcal{M}$, that means

$$P[\mathcal{M}] = P[M] \text{ for all } M \in \mathcal{M}$$

4.

4. (*) On probability density functions

[4 Points]

(a) Let

$$f(x) := \frac{1}{x^k} \mathbb{1}_{[1, +\infty)}(x).$$

For what value of $k \in \mathbb{R}$, if any, is f a probability density function?

4. a.

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{\infty} \frac{1}{x^k} \mathbb{1}_{[1, +\infty)}(x) dx \\
 &= \int_1^{+\infty} \frac{1}{x^k} dx \\
 &= \left[-\frac{1}{k-1} x^{-(k-1)} \right]_1^{+\infty} \\
 &= \frac{1}{k-1}
 \end{aligned}$$

Set $k=2$, then $\int_{-\infty}^{\infty} f(x) = 1$.

Also note $f(x)$ is continuous on $[1, +\infty)$
 so it's integrable by FTC I. } and zero elsewhere

- (b) Give an example of a probability density function f such that $c\sqrt{f}$ cannot be a probability density function for any $c > 0$.

4.b. We can find a function f such that

$$\sqrt{f} \in L_2 \text{ and } \sqrt{f} \notin L_1$$

easier is for some scaling factor k .

$$f(x) = \frac{k}{x^2} \mathbb{1}_{[1, \infty)}(x)$$

$$\begin{aligned} \text{Find } k: \int_{-\infty}^{\infty} f(x) dx &= \int_1^{\infty} \frac{k}{x^2} dx & \text{So } k=1, \\ &= \left[-\frac{k}{x} \right]_1^{\infty} & \text{for } f(x) \text{ to be PDF.} \\ &= k \end{aligned}$$

Also note $f(x)$ is continuous, thus integrable by FTC $\int_1^{\infty}$ on $[1, +\infty)$ and zero elsewhere

$$\begin{aligned} \text{For } \sqrt{f}, \quad \int_{-\infty}^{\infty} \sqrt{f(x)} dx &= \int_1^{\infty} \frac{\sqrt{k}}{x} dx \\ &= +\infty \quad (\text{by the p-test}) \end{aligned}$$

(c) Let

$$f(x) = 2|x| (1 - x^2) \mathbb{1}_{\{|x| \leq 1\}}.$$

i. Find the corresponding cumulative distribution function F .

4.c.i. $f(x)$ is continuous on its support, thus integrable.

$$F(x) = \int_{-\infty}^x f(x) dx = \begin{cases} 0 & x \in (-\infty, -2) \\ \int_{-2}^x (-2x + 2x^3) dx & x \in [-2, 0) \\ \int_{-2}^0 (-2x + 2x^3) dx + \int_0^x (2x - 2x^3) dx & x \in [0, 1) \\ \int_{-2}^0 (-2x + 2x^3) dx + \int_0^1 (2x - 2x^3) dx & x \in [1, +\infty) \end{cases}$$

$$\int_{-2}^x (-2x + 2x^3) dx$$

$$= \left[-x^2 + \frac{1}{2}x^4 \right]_{-2}^x = -x^2 + \frac{1}{2}x^4 + \frac{1}{2} \quad (\text{when } x=0, \frac{1}{2})$$

$$\int_0^x (2x - 2x^3) dx$$

$$= \left[x^2 - \frac{1}{2}x^4 \right]_0^x = x^2 - \frac{1}{2}x^4 \quad (\text{when } x=1, \frac{1}{2})$$

$$\text{So } F(x) = \begin{cases} 0 & x \in (-\infty, -1) \\ \frac{1}{2} - x^2 + \frac{1}{2}x^4 & x \in [-1, 0) \\ \frac{1}{2} + x^2 - \frac{1}{2}x^4 & x \in [0, 1) \\ 1 & x \in [1, +\infty) \end{cases}$$

ii. Let $\mathbf{P}$ be the associated probability measure. Compute $\mathbf{P}[(-\infty, -\frac{1}{2})]$ and $\mathbf{P}[(-\frac{1}{2}, \frac{1}{2})]$.

4. c. ii. Since $F(x)$ has density $f(x)$, so

$$\underbrace{P(X \leq c)}_{F(c)} = P(X < c)$$

$$\begin{aligned} P\left(-\infty, -\frac{1}{2}\right] &= F\left(-\frac{1}{2}\right) - \overbrace{F(-\infty)}^0 \\ &= \frac{1}{2} - \left(-\frac{1}{2}\right)^2 + \frac{1}{2} \left(-\frac{1}{2}\right)^4 \\ &= \frac{1}{2} - \frac{1}{4} + \frac{1}{32} \\ &= \frac{9}{32} \end{aligned}$$

$$\begin{aligned} P\left[-\frac{1}{2}, \frac{1}{2}\right] &= F\left(\frac{1}{2}\right) - F\left(-\frac{1}{2}\right) \\ &= \frac{1}{2} f\left(\frac{1}{2}\right)^2 - \frac{1}{2} \left(-\frac{1}{2}\right)^4 - \frac{9}{32} \\ &= \frac{1}{2} + \frac{1}{4} - \frac{1}{32} - \frac{9}{32} \\ &= \frac{7}{16} \end{aligned}$$